

1 General Notes

1.1 Overview

This document provides general information on the development and configuration of software using the MES Development Suite.

You can use the MDS to customize existing software or to create your own applications.

1.2 Namespaces and scopes

1.2.1 Namespaces

Use namespaces to ensure that software of MPDV and of customers can be installed and operated free of conflicts in the system. Namespaces are implemented using naming conventions of identifiers.

If the customer makes customizations or extensions, the customer must use new identifiers in the customer namespace "u_???".

Category	Namespace	Explanatory notes
Customer	u_???	Names of artifacts, objects or other identifiers, which are created by the customer or specifically for the customer, must start with the prefix "u_". Case-sensitivity is not globally specified. You use upper or lower case letters according to the context. Example of a service name (upper case letter): U_MyObjectName.listObjects Example of a database table (lower case letter): u_myobjectname
MPDV	All other	Artifacts, objects or other identifiers, which are not included in the namespace of customers, are reserved for components of the standard system.

1.2.2 Scopes

The system contains different customization levels (scopes). You can therefore customize standard applications without changing the actual application. With this concept, you store the software artifacts in customization levels (scopes) that overwrite the artifacts of the standard.

The following scopes are provided:

Standard

MPDV delivers the standard applications in the standard scope.

Custom

In the custom scope, MPDV delivers customizations of the standard software in the standard namespace or customer-specific software in the customer namespace "u_???".

Local

Customers use the local scope to customize the applications. In the local scope, you can customize all applications, regardless of the namespace. The end customer is responsible for the content of the local scope.

Software artifacts of the special scopes take priority over the artifacts of the general scopes. The local scope has the highest priority, the standard scope the lowest priority.

Software artifacts of the special scopes override the artifacts of the general scopes. There are different forms of overriding artifacts:

- **Sequential execution:** With program code (e.g. user exits of services), all existing program parts of the different scopes are executed **one after the other**. First the program code of the general scope is executed, then the code of the special scope. In an ideal case, the underlying data structures are built in a way so that the more special scopes can undo actions of the general scopes.
- **Merge:** A file or object of the special scope is added to the object of the general scope. The individual contents of the special scope overwrite and add to the defined contents of the file or the object of the general scope. Example 1: you can add further columns to a service configured in the standard scope. Here, only the additional columns are listed in a configuration file in the custom scope. Example 2: the permitted operators of a service parameter can be overwritten because the service parameter of the general scope is completely overwritten with the contents of the more special scope.
- **Replace:** A file or object of the special scope completely **replaces** the object of the general scope, e.g. configuration files.

1.3 Important notes



Respect the namespaces and scopes.



Only use latin letters, underlines, numbers and dots in identifiers. Do not use a number as first character of an identifier. Only use dots if this is necessary to structure an identifier, e.g. to structure service parameters (e.g. "person.name").

Do not use special characters!

Special characters like hyphen, slant line or umlauts or characters outside of the ASCII character set can lead to errors in the system.



To finally install your extensions, always use an Update Package and the Maintenance Manager. If you have only deployed your extensions via "deploy by copy", the extensions can be deleted in the course of a later update using the Maintenance Manager.